CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 8: **Sound Waves: Characteristics and Applications** | Quick revision, one page

Wave nature of sound

- Sound is produced by vibrating bodies and needs a material medium; it cannot travel through a vacuum.
- In air it is a longitudinal wave: compressions (high pressure) and rarefactions (low pressure) travel along the direction of propagation.
- Wavelength λ: distance between two consecutive compressions (or rarefactions); amplitude: maximum displacement of particles.

Wave relations

- Frequency f (Hz) $=$ number of vibrations per second; time period $T = \dfrac{1}{f}$ (s).
- Wave speed $v = f\lambda$ (m/s).
- Speed of sound: solids $>$ liquids $>$ gases (about 340 m/s in air at room temperature, 1500 m/s in sea water); it increases with temperature in a medium.

Characteristics of sound

- Pitch is determined by frequency (higher frequency, higher pitch); loudness by amplitude (larger amplitude, louder sound); quality (timbre) distinguishes sounds of the same pitch and loudness.
- Intensity is the energy passing per second per unit area; the loudness level is measured in decibel (dB).

Audible range and echo

- Audible range for humans: 20 Hz to 20 000 Hz; infrasound < 20 Hz; ultrasound $> 20\,000$ Hz.
- Echo: reflection of sound. For a distinct echo the time gap must be at least 0.1 s, so the distance of the reflector is at least $\dfrac{v \times 0.1}{2} \approx 17$ m.
- Depth or distance by echo: $d = \dfrac{v\,t}{2}$, where t is the time between sending and receiving the sound.

Applications

- SONAR (Sound Navigation And Ranging): depth of sea, submarines, shoals of fish. Ultrasound: flaw detection, medical scans, cleaning. Echolocation: bats, dolphins.
- Reverberation and noise: excessive loud sound (about 120 dB and above) can damage hearing; reduce by sound-absorbing materials.

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